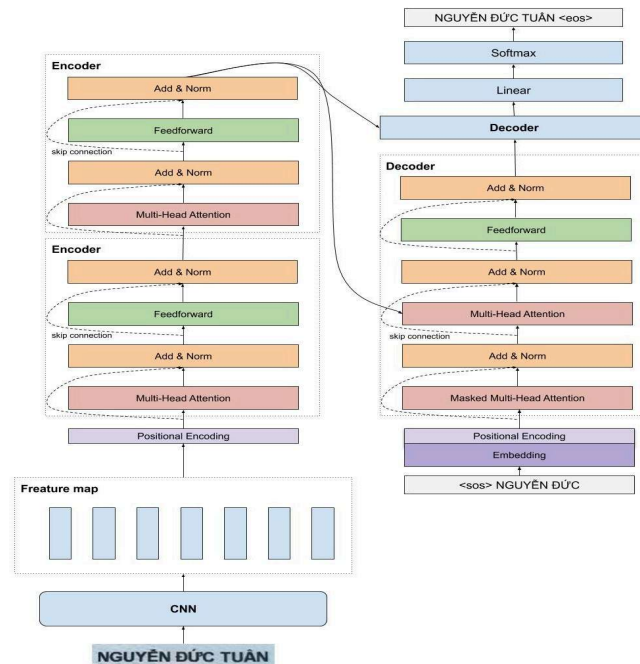


▼ Introduction

- Việc sử dụng [VietOCR](#) rất phù hợp đối với các data custom như chữ trên bìu sách.
- Model sử dụng : TransformerOCR.



- Lý do lựa chọn vì model có độ chính xác cao nhất và phù hợp với data phức tạp như chữ trên bìu sách.
- Dataset:
 - 22000 dòng text được cắt ra từ bìu sách và dán nhãn.
 - 100000 dòng text generate với nhiều font chữ khác nhau.

```
# Cài đặt thư viện
! pip install --quiet vietocr==0.3.5
```

```
61kB 4.8MB/s
952kB 21.4MB/s
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing wheel metadata ... done
286kB 38.8MB/s
Building wheel for gdown (PEP 517) ... done
ERROR: alumentations 0.1.12 has requirement imgaug<0.2.7,>=0.2.5, but you'll have imgaug 0.4.0 which is incompatible.
```

▼ Import thư viện

```
import matplotlib.pyplot as plt
from PIL import Image

from vietocr.tool.predictor import Predictor
from vietocr.tool.config import Cfg
from vietocr.model.trainer import Trainer
```

▼ Download dataset

```
! gdown https://drive.google.com/uc?id=1u0rdBc8phxASIVTpR76v2fmXuXjmVHAG
```

```
Downloading...
From: https://drive.google.com/uc?id=1u0rdBc8phxASIVTpR76v2fmXuXjmVHAG
To: /content/data.zip
1.95GB [00:32, 59.8MB/s]
```

```
! unzip -qq -o ./data.zip
```

✓ Train model

1. Load your config
2. Train model using your dataset above

✓ Thay đổi cấu hình

- *data_root*: thư mục lưu tất cả các ảnh
- *train_annotation*: đường dẫn đến file train
- *valid_annotation*: đường dẫn đến file valid
- *print_every*: in train loss sau n bước
- *valid_every*: in validation loss sau n bước
- *iters*: số vòng để train model của bạn
- *export*: thư mục lưu weight để tham chiếu
- *metrics*: số lượng ví dụ lưu trong validation mà bạn muốn để tính loss theo từng phần, với bộ dữ liệu lớn thì việc chia ra này sẽ giúp tiết kiệm thời gian, không gian bộ nhớ hơn.

Load config mặc định ('vgg_transformer')

```
config = Cfg.load_config_from_name('vgg_transformer')
```

```
dataset_params = {
    'name': 'hw',
    'data_root': './data/',
    'train_annotation': 'train.txt',
    'valid_annotation': 'test.txt'
}

params = {
    'print_every': 200,
    'valid_every': 15 * 200,
    'iters': 10000,
    'checkpoint': './checkpoint/transformerocr_checkpoint.pth',
    'export': './weights/transformerocr.pth',
    'metrics': 10000
}

config['trainer'].update(params)
config['dataset'].update(dataset_params)
config['device'] = 'cuda:0'
```

Bạn có thể thay đổi bất kì tham số ở danh sách bên dưới

```
config
```

```
{'aug': {'image_aug': True, 'masked_language_model': True},
 'backbone': 'vgg19_bn',
 'cnn': {'hidden': 256,
        'ks': [[2, 2], [2, 2], [2, 1], [2, 1], [1, 1]],
        'pretrained': True,
        'ss': [[2, 2], [2, 2], [2, 1], [2, 1], [1, 1]]},
 'dataloader': {'num_workers': 3, 'pin_memory': True},
 'dataset': {'data_root': './data/',
             'image_height': 32,
             'image_max_width': 512,
             'image_min_width': 32,
             'name': 'hw',
             'train_annotation': 'train.txt',
             'valid_annotation': 'test.txt'},
 'device': 'cuda:0',
 'optimizer': {'max_lr': 0.0003, 'pct_start': 0.1},
 'predictor': {'beamsearch': False},
 'pretrain': {'cached': '/tmp/transformerocr.pth',
              'id_or_url': '13327Y1tz1ohsm5YZMyXVMPiOjo0A00aA',
              'md5': 'af6b46e9295eb1c27ca90bddf8c8729a'},
 'quiet': False,
 'seq_modeling': 'transformer',
 'trainer': {'batch_size': 32,
             'checkpoint': './checkpoint/transformerocr_checkpoint.pth',
             'export': './weights/transformerocr.pth',
             'iters': 10000,
             'log': './train.log',
             'metrics': 10000,
             'print_every': 200,
```

[illegible]

- ✓ Bắt đầu training

Tiền xử lý dữ liệu và áp dụng weight có sẵn giúp tăng nhanh thời gian huấn luyện

```
trainer = Trainer(config, pretrained=True)

Downloading: "https://download.pytorch.org/models/vgg19\_bn-c79401a0.pth" to /root/.cache/torch/hub/checkpoints/vgg19_bn-c79401a0.pth
100%          548M/548M [00:29<00:00, 19.4MB/s]

Cached Downloading: /tmp/tranformerorc.pth
Downloading...
From: https://drive.google.com/uc?id=13327Y1tz1ohsm5YZMyXVMPIOjoOA00aA
To: /root/.cache/gdown/tmp94iadf37/dl
152MB [00:01, 97.9MB/s]
Computing MD5: /tmp/tranformerorc.pth
MD5 matches: /tmp/tranformerorc.pth
Create train_hw: 100%|██████████| 121868/121868 [02:11<00:00, 927.14it/s]
Created dataset with 121867 samples
train_hw build cluster: 100%|██████████| 121867/121867 [00:01<00:00, 87007.58it/s]
/usr/local/lib/python3.7/dist-packages/torch/utils/data/dataloader.py:481: UserWarning: This DataLoader will create 3 worker
cpuset_checked))
Create valid_hw: 100%|██████████| 22637/22637 [00:32<00:00, 705.58it/s]
Created dataset with 22636 samples
valid_hw build cluster: 100%|██████████| 22636/22636 [00:00<00:00, 83604.57it/s]
```

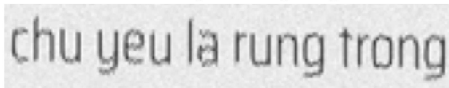
Visualize dataset để đối chiếu

```
/usr/local/lib/python3.7/dist-packages/torch/utils/data/dataloader.py:481: UserWarning: This DataLoader will create 3 worker processes on this machine. This is not a problem if you have sufficient free memory on your machine. If you are using a machine with little available RAM, you may get an OOM error. To fix that, you can set the environment variable 'MALLOC_ARENA_MAX=1' if you are using Linux. Please refer to the man page of malloc(1) for more information. In MacOS, you should try to set the environment variable 'malloc_jit_enabled=0' to fix it. (cpuset_checked))
```

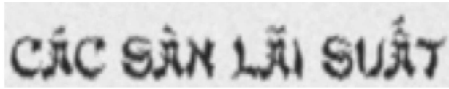
sent: KHOA *QC



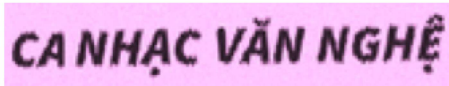
sent: chu yeu la *ung trong



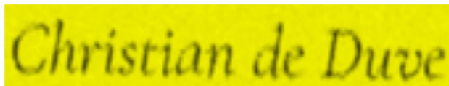
sent: CÁC SẢN L*Ị SUẤT



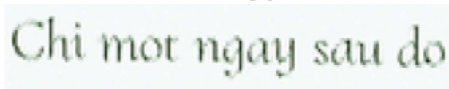
sent: CA NHẠC VĂN NG*Ệ



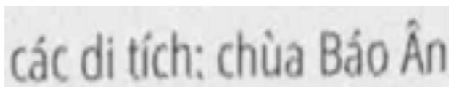
sent: *hristian de Duve



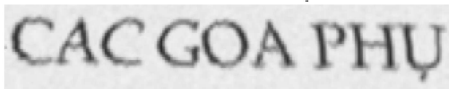
sent: Chi mot*ngay sau do



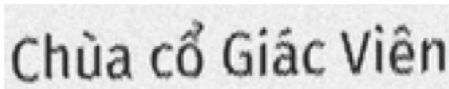
sent: các di tích: chùa Báo Ân



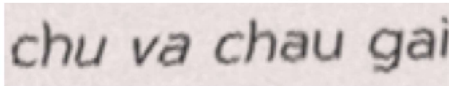
sent: CÁC GÓA*PHỤ



sent: Chùa cổ Giác Viên



sent: chu v* chau gai



sent: K*oi nghi*n



Bắt đầu training

```
trainer.train()
```

```
/usr/local/lib/python3.7/dist-packages/torch/utils/data/dataloader.py:481: UserWarning: This DataLoader will create 3 worker processes on this machine. This is not a problem if you have sufficient free memory on your machine. If you are using a machine with little available RAM, you may get an OOM error. To fix that, you can set the environment variable 'MALLOC_ARENA_MAX=1' if you are using Linux. Please refer to the man page of malloc(1) for more information. In MacOS, you should try to set the environment variable 'malloc_jit_enabled=0' to fix it. (cpuset_checked))
```

```
iter: 000200 - train loss: 0.652 - lr: 3.96e-05 - load time: 2.06 - gpu time: 189.94
iter: 000400 - train loss: 0.589 - lr: 1.12e-04 - load time: 0.23 - gpu time: 186.25
iter: 000600 - train loss: 0.585 - lr: 2.01e-04 - load time: 0.20 - gpu time: 181.44
iter: 000800 - train loss: 0.578 - lr: 2.73e-04 - load time: 0.27 - gpu time: 188.44
iter: 001000 - train loss: 0.577 - lr: 3.46e-04 - load time: 0.30 - gpu time: 183.15
iter: 001200 - train loss: 0.592 - lr: 3.00e-04 - load time: 0.29 - gpu time: 187.30
iter: 001400 - train loss: 0.585 - lr: 2.99e-04 - load time: 0.20 - gpu time: 186.36
iter: 001600 - train loss: 0.570 - lr: 2.97e-04 - load time: 0.22 - gpu time: 185.29
iter: 001800 - train loss: 0.570 - lr: 2.97e-04 - load time: 0.25 - gpu time: 176.84
iter: 002000 - train loss: 0.571 - lr: 2.97e-04 - load time: 0.28 - gpu time: 196.92
iter: 002200 - train loss: 0.571 - lr: 2.97e-04 - load time: 0.25 - gpu time: 186.55
iter: 002400 - train loss: 0.582 - lr: 2.82e-04 - load time: 0.25 - gpu time: 178.41
iter: 002600 - train loss: 0.573 - lr: 2.77e-04 - load time: 0.35 - gpu time: 183.06
iter: 002800 - train loss: 0.571 - lr: 2.71e-04 - load time: 0.18 - gpu time: 179.88
iter: 003000 - train loss: 0.592 - lr: 2.65e-04 - load time: 0.27 - gpu time: 183.69
iter: 003200 - valid loss: 0.571 - acc: 0.7519 - acc per char: 0.9290
iter: 003400 - train loss: 0.563 - lr: 2.58e-04 - load time: 0.31 - gpu time: 181.83
iter: 003600 - train loss: 0.572 - lr: 2.50e-04 - load time: 0.28 - gpu time: 183.02
iter: 003800 - train loss: 0.577 - lr: 2.42e-04 - load time: 0.27 - gpu time: 188.14
iter: 004000 - train loss: 0.571 - lr: 2.34e-04 - load time: 3.35 - gpu time: 175.89
iter: 004200 - train loss: 0.570 - lr: 2.25e-04 - load time: 0.26 - gpu time: 181.27
iter: 004400 - train loss: 0.555 - lr: 2.16e-04 - load time: 0.30 - gpu time: 179.24
iter: 004600 - train loss: 0.559 - lr: 2.06e-04 - load time: 0.25 - gpu time: 181.56
iter: 004800 - train loss: 0.564 - lr: 1.96e-04 - load time: 0.26 - gpu time: 188.24
```

```
iter: 004800 - train loss: 0.562 - lr: 1.86e-04 - load time: 0.22 - gpu time: 183.96
iter: 005000 - train loss: 0.555 - lr: 1.76e-04 - load time: 0.32 - gpu time: 188.42
iter: 005200 - train loss: 0.554 - lr: 1.66e-04 - load time: 0.33 - gpu time: 177.76
iter: 005400 - train loss: 0.556 - lr: 1.55e-04 - load time: 0.24 - gpu time: 180.89
iter: 005600 - train loss: 0.563 - lr: 1.45e-04 - load time: 0.28 - gpu time: 183.50
iter: 005800 - train loss: 0.560 - lr: 1.34e-04 - load time: 0.25 - gpu time: 191.21
iter: 006000 - train loss: 0.560 - lr: 1.24e-04 - load time: 0.30 - gpu time: 185.34
iter: 006000 - valid loss: 0.549 - acc full seq: 0.8176 - acc per char: 0.9533
iter: 006200 - train loss: 0.554 - lr: 1.14e-04 - load time: 0.19 - gpu time: 182.73
iter: 006400 - train loss: 0.557 - lr: 1.04e-04 - load time: 0.25 - gpu time: 189.62
iter: 006600 - train loss: 0.554 - lr: 9.38e-05 - load time: 0.20 - gpu time: 178.57
iter: 006800 - train loss: 0.552 - lr: 8.42e-05 - load time: 0.24 - gpu time: 183.22
iter: 007000 - train loss: 0.548 - lr: 7.50e-05 - load time: 0.23 - gpu time: 189.62
iter: 007200 - train loss: 0.551 - lr: 6.61e-05 - load time: 0.17 - gpu time: 188.07
iter: 007400 - train loss: 0.553 - lr: 5.76e-05 - load time: 0.24 - gpu time: 188.84
iter: 007600 - train loss: 0.551 - lr: 4.96e-05 - load time: 2.41 - gpu time: 184.62
iter: 007800 - train loss: 0.556 - lr: 4.21e-05 - load time: 0.26 - gpu time: 184.09
iter: 008000 - train loss: 0.554 - lr: 3.51e-05 - load time: 0.25 - gpu time: 186.32
iter: 008200 - train loss: 0.548 - lr: 2.86e-05 - load time: 0.22 - gpu time: 183.95
iter: 008400 - train loss: 0.540 - lr: 2.28e-05 - load time: 0.26 - gpu time: 184.26
iter: 008600 - train loss: 0.545 - lr: 1.75e-05 - load time: 0.23 - gpu time: 174.11
iter: 008800 - train loss: 0.539 - lr: 1.29e-05 - load time: 0.27 - gpu time: 180.77
iter: 009000 - train loss: 0.535 - lr: 9.03e-06 - load time: 0.34 - gpu time: 186.28
iter: 009000 - valid loss: 0.538 - acc full seq: 0.8577 - acc per char: 0.9643
iter: 009200 - train loss: 0.541 - lr: 5.80e-06 - load time: 0.26 - gpu time: 188.95
iter: 009400 - train loss: 0.540 - lr: 3.27e-06 - load time: 0.26 - gpu time: 184.70
iter: 009600 - train loss: 0.548 - lr: 1.45e-06 - load time: 0.25 - gpu time: 185.89
iter: 009800 - train loss: 0.542 - lr: 3.63e-07 - load time: 0.26 - gpu time: 179.79
iter: 010000 - train loss: 0.553 - lr: 1.21e-09 - load time: 0.24 - gpu time: 191.75
```

Visualize prediction

```
trainer.visualize_prediction()
```

```
/usr/local/lib/python3.7/dist-packages/torch/utils/data/dataloader.py:481: UserWarning: This DataLoader will create 3 work
cpuset_checked))
```

prob: 0.931 - pred: SKE - actual: SKE



prob: 0.893 - pred: Nước - actual: Nước



prob: 0.940 - pred: ĐỪNG - actual: ĐỪNG



prob: 0.829 - pred: Cùng - actual: cùng



prob: 0.932 - pred: NHỈ? - actual: NHỈ?

